

=====

PhD Qualifying Exam

Topic: Quantum Architecture - Locality Improvement Technologies

Will Buziak

2026 Fall

=====

On real quantum hardware, two qubits can interact directly only when the hardware supports that connection. Otherwise, the compiler or the hardware must move quantum information or create a remote connection. This exam studies ways to improve locality, meaning ways to keep required operations close to the qubits that perform them. Compare how these methods affect execution time, error, circuit depth, and the amount of hardware they require.

Prepare a comprehensive survey paper on this topic. Your survey should answer the following questions and organize them into well-structured sections:

Locality Basics in Quantum Computers:

- How can a program's required qubit interactions be compared with the hardware's physical connection map?
- Why are qubit placement and routing needed when the hardware has limited connections?

Coordinating Hardware and Compiler Design:

- How do the useful locality methods differ for different kinds of machines?
- What programs, hardware connection maps, performance measures, and parameter studies are needed for a fair comparison?

Limits and Open Questions:

- What makes remote entanglement, qubit movement, changing connections, calibration, scheduling, and classical control difficult to scale?
- How should an evaluation handle unwanted effects between nearby operations, often called crosstalk, and errors that affect several operations at once? How can researchers check whether a simplified error model matches real hardware?

Your survey should:

- Be 6 pages long excluding references and follow the two-column ACM conference format.
- Include a comprehensive reference list with at least 20 peer-reviewed papers.

- Compare the papers: explain their assumptions, how the proposed systems work, what evidence supports their claims, their limitations, and the questions that remain open. Do not simply summarize one paper at a time.

Oral Presentation

Prepare a 30-minute oral presentation summarizing the findings of your literature survey. Be ready to address the following during the Q&A session:

- Why locality is a problem in quantum computers and how hardware and compilers address it
- The strengths, costs, and conditions under which each locality-improvement method is useful
- The most important limitations and open design questions identified in your survey.

This presentation will be evaluated on clarity, technical understanding, use of evidence, and ability to explain and defend your conclusions.

Deliverables

- A 6-page survey following the two-column ACM conference format and addressing the outlined questions. (Due: 11:59 pm MT on Dec. 7, 2026)
- A 30-minute oral presentation with slides summarizing your findings. Please schedule this to occur in the final exam week.

Best of luck on your qualifying exam!